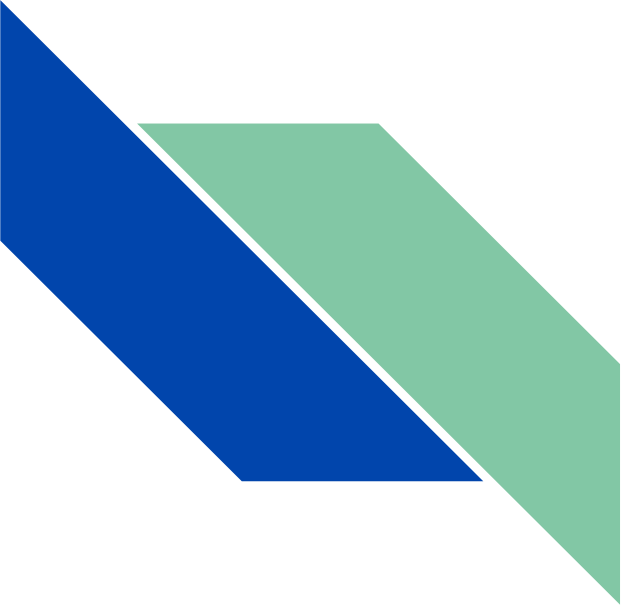




AI Doctor Assistant

(With vision and Voice)

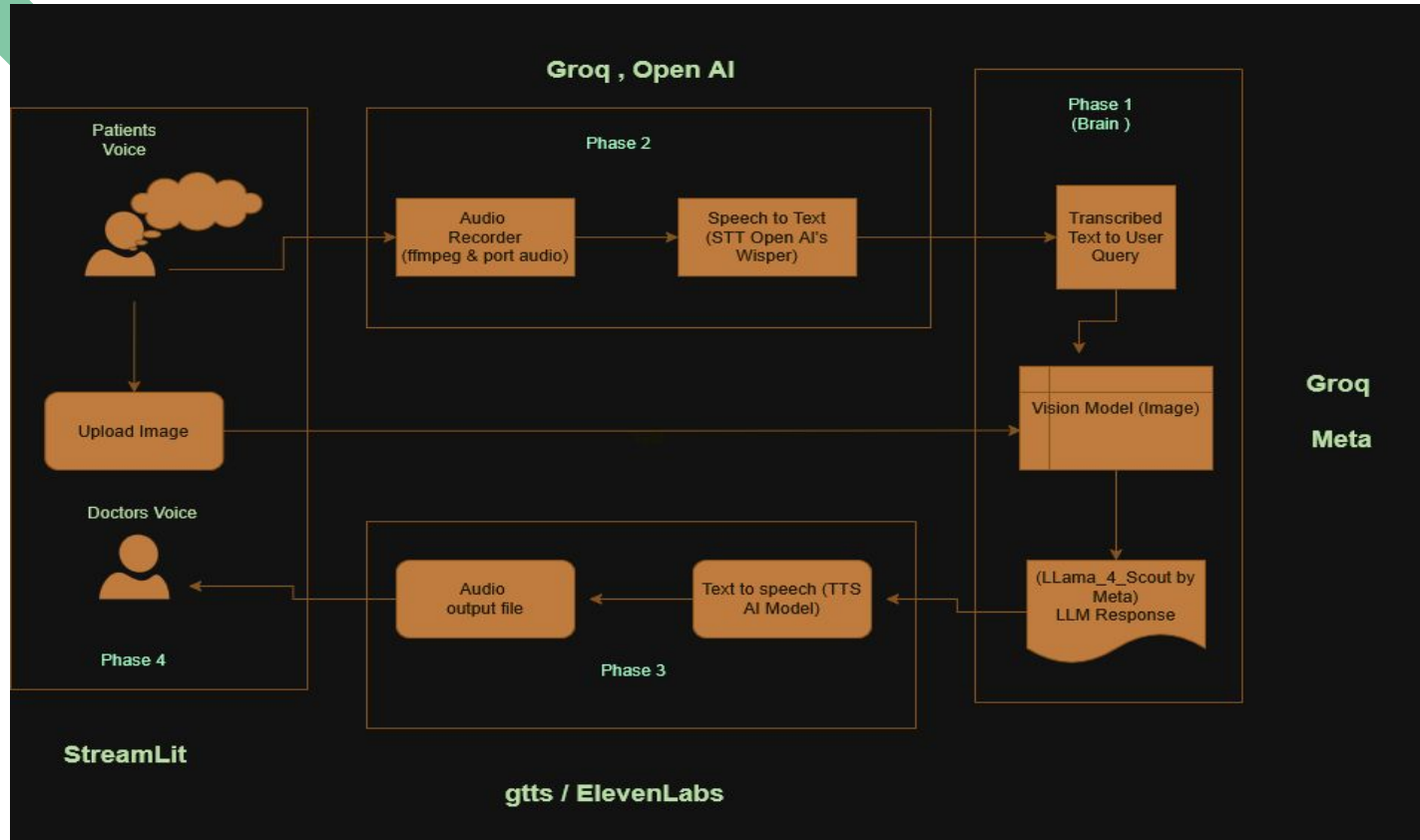
Medical chatbot with Multimodel LLM



Use case..

An AI-powered assistant designed to offer preliminary evaluation of skin-related concerns for patients (not a medical diagnosis) while assisting doctors /dermatologists as a virtual assistant when they are unavailable.

TECHNICAL ARCHITECTURE





TOOLS AND TECHNOLOGIES

- Groq for AI Inference
- OpenAI Whisper (Best open source model for transcription - speech to text)
- Llama 4 Scout Vision (Open source by Meta - multimodal - fast and smart, doesn't waste brainpower)
- gTTS (Googles Text to speech) / ElevenLabs
- Langchain (monitoring and debugging)
- StreamLit for UI
- Python



Challenges faced..

- Ffmpeg needs to be installed to make it work so containerized the app
- Used StreamLit GUI to run the containerized app but was facing issues while building.
- But app worked fine with render.com
- When adding Langsmith (for monitoring) the app worked on the local machine but the containerized app was not working. Still needs to be debugged



Future Enhancements

- Leverage Retrieval-Augmented Generation (RAG) to connect directly to authoritative dermatologist documents, ensuring responses are accurate and grounded in verified medical knowledge.
- Transform the solution into an AI agent capable of autonomously performing tasks using integrated tools such as image analysis and text analysis with results routed to the LLM
- Use state of the art Paid LLM's especially for vision
- Conduct Evaluations using the provided dermatologist reference documents to validate accuracy and reliability.
- Enable multilingual support to make the solution accessible to a broader, global audience.



Thank you